The probabilistic continuous Hilbert transform has L^2 norm strictly larger than one

Autonomous counterexample packet for arXiv:2209.09737

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Abstract

The source predicts that its probabilistic continuous Hilbert transform T has the classical sharp L^p norm $\cot(\pi/(2p^*))$. The conjecture is already false for $p = 2$. The source kernel is the Hilbert kernel plus an odd integrable correction J which is strictly positive on $(1, \infty)$. Its finite positive first half-moment implies $\widehat{J}(\xi) = -2iM\xi + o(\xi)$ as $\xi \downarrow 0$. This reinforces the Hilbert multiplier $-i$, giving multiplier modulus strictly larger than one at small positive frequencies.

1 Source conjecture and kernel

The source's Corollary 9.3 (PDF page 52) writes the kernel of the probabilistic continuous Hilbert transform as

$$K_T(x) = \frac{1}{\pi x} + J(x), \tag{1}$$

where

$$J(x) = \frac{\mathbf{1}_{\{|x|\geq 1\}}}{\pi x} \int_0^\infty \frac{2y^3}{(y^2 + \pi^2 x^2)\sinh^2 y}\, dy. \tag{2}$$

It proves $J \in L^1(\mathbb{R})$ and obtains

$$\cot\left(\frac{\pi}{2p^*}\right) \leq \|T\|_{L^p \to L^p} \leq \cot\left(\frac{\pi}{2p^*}\right) + \|J\|_1.$$

Conjecture 9.6 (PDF page 53) asks whether the lower bound is always an equality:

> **Conjecture 9.6.** *We conjecture that the L^p-norm of the operator T should be $\cot(\pi/(2p^*))$ and similarly for T^k. For the latter, even the conjecture that the norm is independent of d would be of interest. See the discussion preceding (6.13).*

2 The correction has a positive first half-moment

The function J is real and odd. Moreover, $J(x) > 0$ for every $x > 1$. Set

$$M := \int_1^\infty xJ(x)\, dx. \tag{3}$$

Theorem 1. *One has $0 < M < \infty$.*

Proof. Positivity follows directly from (2). For $x \geq 1$,

$$0 < J(x) \leq \frac{1}{\pi^3 x^3} \int_0^\infty \frac{2y^3}{\sinh^2 y}\, dy.$$

The y-integral is finite: near zero its integrand is $O(y)$, and at infinity it decays exponentially. Hence $xJ(x) \leq Cx^{-2}$ on $[1, \infty)$, proving finiteness of M; strict positivity gives $M > 0$. □

3 The L^2 multiplier

Use the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-ix\xi}\, dx$. The principal-value kernel $1/(\pi x)$ has multiplier $-i \operatorname{sgn} \xi$. Because J is real and odd,

$$\widehat{J}(\xi) = -2i \int_1^\infty J(x) \sin(x\xi)\, dx =: -ia(\xi). \tag{4}$$

Theorem 1 permits dominated convergence after division by ξ:

$$\lim_{\xi \downarrow 0} \frac{a(\xi)}{\xi} = 2 \int_1^\infty xJ(x)\, dx = 2M > 0. \tag{5}$$

Thus $a(\xi) > 0$ throughout some interval $(0, \delta)$.

Theorem 2. *For the operator T of the source,*

$$\|T\|_{L^2(\mathbb{R}) \to L^2(\mathbb{R})} > 1.$$

Consequently Conjecture 9.6 is false.

Proof. By (1) and (4), the Fourier multiplier of T at positive frequencies is

$$m_T(\xi) = -i + \widehat{J}(\xi) = -i(1 + a(\xi)).$$

For $0 < \xi < \delta$, (5) gives $|m_T(\xi)| = 1 + a(\xi) > 1$. The function a is continuous, since $J \in L^1$, so the strict inequality holds on a set of positive measure. Plancherel's theorem yields

$$\|T\|_{2 \to 2} = \operatorname*{ess\,sup}_{\xi \in \mathbb{R}} |m_T(\xi)| > 1.$$

For $p = 2$ one has $p^* = 2$ and $\cot(\pi/(2p^*)) = \cot(\pi/4) = 1$, contradicting the conjectured identity. □

4 Scope

This disproves the assertion about the one-dimensional operator T; therefore the combined Conjecture 9.6 is false. It does not determine the exact L^2 norm, nor does it settle the separate dimension-free or sharp-norm questions for the higher-dimensional operators $T^{(k)}$ or for the classical discrete Riesz transforms.

References

[1] R. Bañuelos, D. Kim, and M. Kwaśnicki, *Sharp ℓ^p inequalities for discrete singular integrals on the lattice $\mathbb{Z}^d$*, arXiv:2209.09737, Corollary 9.3, Theorem 9.5, and Conjecture 9.6.